Result 1 (binary module intersection membership). [Topology foundations, submodule intersection]
Let K be a type. Let *field* be a field structure on K. Let M be a module over K. Let U and V be elements of Subspace(M). Let x be an element of M. Then

$$x \text{ belongs to } (U \cap V) \text{ if and only if } x \text{ belongs to } U \text{ and } x \text{ belongs to } V.$$

Result 2 (extreme perturbations have certificates). [Bendich, Edelsbrunner, Morozov, and Patel, Theorem 4 extreme perturbations]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume P.interval.lower $\leq P$.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Assume $X_r(P)$ is a nested pair (`nesting`). Then

$$(\text{there exists } E : \mathrm{Ext}_H(P, r),\ E.\text{top.data.map} = (P.map - (r)) \text{ and } E.\text{bottom.data.map} = (P.map - (-r))).$$

Adapted from the cited source. Scope: existence of certificates.

Result 3 (extreme image certificate independence). [Topology foundations, two-image certificate independence]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume $X_r(P)$ is a nested pair (`nesting`). Let E and F be elements of $\mathrm{Ext}_H(P, r)$. Then

$$I_r^+(P; H, E) \cap I_r^-(P; H, E) = I_r^+(P; H, F) \cap I_r^-(P; H, F).$$

Result 4 (well group lies in both extreme images). [Bendich, Edelsbrunner, Morozov, and Patel, Theorem 4]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let r be a real number. Assume $X_r(P)$ is a nested pair (`nesting`). Let E be certificates for the two extreme perturbations. Then

$$U_r(P; H) \text{ is contained in } I_r^+(P; H, E) \cap I_r^-(P; H, E).$$

Adapted from the cited source. Scope: forward inclusion only.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.